Chapter 1

Introduction

1.1 Motivation

When several quadrotors fly close together, the downwash of the upper vehicle can destabilise the one below. The usual remedy is to stay far enough apart that the interaction can be ignored. For a Crazyflie 2.0 (9 cm rotor-to-rotor) that conservative planning gap has been taken as about 60 cm; Shi et al. [7] cite that figure from earlier planning work as one such example.

That margin is expensive. It sets the density at which a team can operate, and therefore the size of the space a given number of vehicles requires. Applications that motivate multirotor teams in the first place — inspection in confined spaces, cooperative transport, aerial docking — are precisely those in which the space is not available. In some of them, close approach is the task rather than a hazard to be avoided.

Removing the margin requires a controller that accounts for the interaction rather than avoiding it, and that is difficult for a specific reason. The interaction is not a small perturbation around a known model: it is strongly nonlinear in the relative configuration of the team, growing sharply as vehicles close and decaying quickly with lateral offset, and it is markedly asymmetric, since the vehicle above is affected far less than the one below. It is also hard to derive from first principles in the regime that matters. Simplified models exist for part of the problem: the mean flow from a quadrotor's propellers in hover is well approximated by a turbulent jet beyond about two and a half motor-to-motor distances below the vehicle [2]. Formation flight happens closer than that. In that near field those authors report no closed-form model.

The field's response has been to obtain the interaction from data rather than from theory, and it has worked. Reported results include a reduction in worst-case height error by a factor of two or better, up to four [7], sixteen vehicles flying at a minimum vertical separation of 24 cm [6], and average reductions of 56 % in vertical and 36 % in three-dimensional tracking error for a pair of vehicles [8]. The margin has been substantially reduced.

1.2 Problem

A controller can obtain information about the interaction force in exactly two ways, and the distinction organises the entire field.

It can **measure** the residual online, by comparing the acceleration the vehicle actually experienced against the acceleration its nominal model predicts for the actuation it applied. This requires no prior knowledge of the interaction and works equally against disturbances

never seen before. But the estimate exists only after the disturbance has begun to act, and in its established form it depends on rotor-speed sensing.

Or it can **predict** the residual before it acts, from the relative states of its neighbours, using a model learned offline. This can compensate ahead of the disturbance rather than after it, but the model is only valid over the configurations its training data covered, and outside them its behaviour is not merely degraded but unconstrained.

The two sources are complementary. Measurement is late but not tied to a training set. Prediction can act earlier but only where it was trained. Combining them is the obvious next construction. It has been built, on a single vehicle. There the authors describe the extra gain over the reactive method as small without a payload, and as leaving performance similar to the reactive method when a payload is attached [3].

The practical question a system designer faces is which to use, and it has no answer in the literature. Not because the individual results are weak, and not because the field avoids systematic comparison. Recent work benchmarks online-learning estimators on a single vehicle [5] and fractional-order and classical regulators for close-proximity swarms [1]. It is that the results for *these* families cannot be composed: each is reported against its own baseline, on its own airframe, at its own separations, on its own trajectories, under its own metrics.

1.3 Gap

Representative methods exist for reactive compensation, for predictive compensation, and for their hybrid, and each is individually validated, but, to the best of our knowledge, they have not been evaluated against one another under identical conditions, and the field's reporting conventions make such a comparison impossible to assemble after the fact. A 36 % improvement measured on two vehicles in close proximity and a 31 % improvement measured in a tight formation are not the same quantity; neither can be subtracted from the other, and their ordering says nothing about which method would prevail if both were flown under the same conditions.

This is not a criticism of any individual work. Each is internally valid, and none violated a convention of the field. It is a property of the literature taken as a whole: the results were not designed to compose, and establishing a ranking requires an experiment built for that purpose.

Two narrower gaps follow. The hybrid has been demonstrated only where the residual is *self-induced*, a single vehicle, with effects such as blade flapping, drag and ground forces, optionally carrying a slung payload, and not where it is produced by other vehicles and is therefore a function of a measurable relative state, which is where the predictive component has the most to contribute; multi-robot evaluation of such hybrids is not in the scope of this thesis. And the summation that predictive architectures use to aggregate per-neighbour contributions is known to be an approximation [7, 4], yet the size of the penalty it carries for a deployed model is not reported.

1.4 Research questions

RQ1 Under identical hardware, trajectories, formations and separations, how do reactive and predictive interaction-force compensation strategies compare to the uncompensated baseline in trajectory-tracking accuracy and residual-force rejection?

RQ2 Does the relative advantage of reactive versus predictive compensation depend on

the interaction regime, separation, relative velocity, and whether the disturbance is quasi-static or transient?

RQ3 How large is the transfer penalty when a summed per-neighbour residual model is trained only on two-vehicle data and then used on three-vehicle formations, and is that penalty small enough that the summed architecture remains the right engineering choice? This measures the size of a known approximation. It is not a test of whether superposition holds: it does not, and that is already established [7, 4]. The question applies to the predictive strategies (Strategy 2 and, if implemented, Strategy 3). It does not apply to a purely reactive law.

RQ4 For each compared strategy, what sensing, onboard compute and training data does it require, and how do those costs sit next to the tracking and rejection it delivers?

RQ1 is the primary question and the one the experimental design serves. RQ2 carries the most engineering value, because it converts a ranking into a rule about when to use which. RQ3 asks for the magnitude of the transfer penalty on a deployed model, not whether superposition holds. RQ4 compares cost against performance for every compared strategy.

The questions are stated with falsifiable hypotheses in Chapter **??** and Chapter **??**; each is answerable by the experiments described in Chapter **??**, and a negative answer to any of them is reported as a result rather than as a failure.

1.5 Contributions

This thesis sets out to establish the following. This list is the intended contribution, not a report of measured outcomes. The experiments that would support it are specified in Chapter **??** and, once flown, belong in the results chapters.

1. **A controlled comparison.** To the best of our knowledge, the first evaluation of reactive and predictive compensation on the same airframe, the same trajectories and the same formations, with one definition of the residual logged under every controller, including those that do not use it, and with null flights that establish the measurement floor. "Controlled" means the experiment is held fixed, not that every law is given the same gains. Geometric tracking, INDI and feedback linearisation are different controllers; each is tuned for itself. Gains are then frozen before the comparison flights so a later difference is not mid-campaign retuning.

2. **A quantification of what the summed per-neighbour approximation costs** when a model trained on two vehicles meets three. This measures a penalty whose *existence* is already established in the literature [7, 4]; the contribution is its magnitude for a deployed model, not the finding that it exists.

3. **An engineering account of cost**: sensing prerequisites, onboard computation and training-data requirements, reported alongside performance so that the comparison supports a decision rather than only a ranking.

Supporting these, but explicitly *not* a scientific contribution, is the apparatus that makes them feasible: the frozen scenario library, the residual measurement path, an onboard network with a weight-upload protocol, an offline training pipeline verified against the compiled controller, and a simulation environment running the same controller source that flies. It is described in Chapter **??** and offered for reuse.

What this thesis does not claim. No new control method is proposed: every strategy follows published work, and the contribution is the comparison. The residual model is not offered as an improvement on published architectures. Results concern a single airframe and a single class of disturbance; generalisation beyond either is discussed rather than demonstrated. No claim rests on simulation, which is used to disprove implementations and never to validate a method. And no strategy is claimed to be best in general, the useful outcome is conditional, which is RQ2.

A negative result would also be a contribution. If the strategies do not separate, the conclusion is that the choice of compensation family matters less than whether compensation is present at all, and the engineering question becomes one of cost. The experimental design is built so that this outcome is distinguishable from a failed experiment, which is what the null conditions and gains frozen after each strategy is tuned are for.

Provenance, not pending work. "To the best of our knowledge" is supported by two audits: an arXiv keyword sweep and a cross-database sweep including a citation-graph scan. Both found adjacent comparisons, cited in §**??**, but none spanning these families. Neither audit is exhaustive.

1.6 Scope

The scope is deliberately narrow, because the contribution is a controlled comparison and every degree of freedom left open is an alternative explanation for an observed difference.

Table 1.1: Scope of this thesis.

In scope	Out of scope
Inter-vehicle aerodynamic interaction, principally downwash	Wind, gusts, and other external disturbances
Teams of two and three vehicles	Large swarms
Homogeneous vehicles	Heterogeneous teams
Tracking of pre-planned, collision-free formations	Interaction-aware planning; collision avoidance
Onboard inference, offline training	Online or on-board learning
The force residual	The moment residual, which is measured but not separately compensated

1.7 Outline

Chapter ?? surveys the field as an argument rather than a catalogue: what the disturbance is, the two ways of obtaining information about it, why the injection point into the control law matters, and why the individual results do not compose into a comparison.

Chapter ?? defines the residual this stack measures and logs. Strategies that consume a force residual use that number. It gives both routes to it in one notation, and states the assumptions the formulation makes together with where each is expected to fail.

Chapter ?? describes Strategies 0–3 (the uncompensated baseline, pure INDI, geometric+NN and FBL+NN), and the point at which each injects its knowledge of the interaction, separating in each case what is implemented here from the published method it follows.

Chapter ?? describes the platform, the frozen scenario library, the measurement chain and the protocol, and states the threats to validity the design is built against, including those it does not eliminate.

The two- and multi-robot results chapters report the two- and three-vehicle results. **The discussion chapter** interprets them against the research questions and the stated assumptions, and **the concluding chapter** concludes.

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